

Sentiment backtest — does hot/cold × bull/bear lead the tape?

Capstone US Equities · internal · 02 September 2026 · universe of 99 wiki names · prepared from the sentiment dashboard build (build_sentiment.py). Every figure below is reproducible from `_wiki/_data/sentiment/`.

1. Bottom line

Sentiment is coincident to lagging, not leading. Twitter and news tone move with the same week's return and after it; they carry almost no information about the following week. The only signals with forward information in this universe are analyst EPS revisions (a small, well-known residual that survives a momentum control) and short interest as a slow positioning level. Sentiment does earn a place in one setting: positioning into a known catalyst. Pre-print tone and attention predict the earnings reaction and the 15 days after it, and being 'hot' into a print raises the bar for the result.

- **Tone is coincident.** Bloomberg Twitter sentiment: rank IC +0.265 with the same week's relative return, +0.128 with the prior week, +0.008 with the next week. News sentiment: +0.358 same week. The reverse test (this week's return → next week's tone) is +0.128: price moves first, the crowd talks afterwards. Identical shape in 2024, 2025 and 2026.
- **Attention is a fade, not a lead.** All-of-X tweet volume vs its own trailing month: IC +0.111 same week, -0.004 next week. As a weekly top-5 basket the hottest names returned -0.07% the following week and the coldest -0.14%.
- **EPS revisions are the one residual.** Mostly lagging (IC ≈ +0.10 on the four prior weeks: revisions chase price) but with a genuine forward tail: +0.039 next week, +0.034 after removing the past-4-week return (t 2.4), persisting to four weeks out. Top-5 basket +0.88%/week vs bottom-5 +0.10%, spread positive in 55% of weeks, +124% cumulative over 138 weeks.
- **Short interest is a level signal.** Flat week-to-week correlations, but inside every momentum tercile the high-SI third beat the low-SI third by +1.3% to +1.8% over 15 days (2025–26 squeeze regime). Top-5 basket +0.71%/week, spread +0.88%, +103% cumulative.
- **Where sentiment does work: into earnings.** Across 1,034 prints, pre-print Twitter sentiment has IC +0.069 on the 15-day return (top-minus-bottom quintile +3.9%); news sentiment +0.093 on the reaction. Hot-into-print names that beat earned +0.4% over 15 days vs +1.7% for cold names that beat; hot names that printed in line lost -5.0%. Post-print sentiment shifts confirm the move (IC +0.424 on the reaction) and do not predict the drift (-0.009).
- **Own corpus and sell-side: too short to conclude.** The curated-handle corpus (May-2026→, 18 weeks) and the Outlook sell-side leg (Feb-2026→) show nothing with $|t| \geq 2$ on the weekly test. An earlier read that curated attention leads (15-day IC +0.04) does not survive the non-overlapping weekly test and should be treated as noise. Sell-side note flow is contrarian; pre-print broker upgrades are the most contrarian pre-print signal in the study (IC -0.122 on the reaction, n=228).

Practical implication: keep the hot/cold ranking as a read on where the crowd is, not as a buy list. Put forecasting weight on revisions, short interest and the pre-print setup; show tone as context.

2. Data and construction

Source	Coverage	What it contributes
Own tweet corpus (twitter-briefing routine, twitterapi.io)	109,232 tweets scanned since 2026-05-01, 15,112 mapped to a ticker; ~440 curated handles	Curated attention (mentions vs own trailing 4 weeks, wires separate) and lexicon tone
Bloomberg social/news fields (bdh)	Daily Twitter sentiment, tweet count, pos/neg counts, news sentiment and count, all 99 names, Dec-2023 → today	All-of-X attention and tone — the long backward track (139 weeks)
Bloomberg estimates/positioning	BEST_EPS (1BF) daily, BEST_ANALYST_RATING, SI_PERCENT_EQUITY_FLOAT, PX_LAST	EPS revision (4-week %), rating drift, short interest level, returns
Outlook sell-side e-mails (COM, DASL sweep)	53,042 broker e-mails since 2026-01-30, 10,047 mapped to 97 tickers (subject + body)	Note flow, note tone, rating changes, price-target changes, estimate direction
Bloomberg earnings history (bds)	1,034 prints with a full 15-day window, Jan-2024 → Aug-2026, time-of-day aware	Event windows: reaction, 15-day total, drift; EPS surprise

- **Construction.** Signals are stamped every Friday and z-scored across names within the week (clipped ± 3). Attention = $\log((\text{count this week} + 0.5) / (\text{trailing 4-week mean} + 0.5))$. Tone = count- or engagement-weighted weekly mean, shrunk toward zero when few observations. HOT = mean z of the attention legs; BULL = mean z of tone, tone change, EPS revision, rating drift (plus sell-side tone and PT change on the own-corpus track).
- **Returns.** Friday-to-Friday closes in local currency; every test uses the return relative to the equal-weight universe, so market beta is removed. Forward horizons 1 week, 15 calendar days, 4 weeks.
- **Tests.** (a) Weekly rank IC (Spearman across names) of each signal vs forward relative return, with quintile spreads and by-year splits. (b) Cross-correlation function: IC of the signal at week t with the return of week t+k, k = -4...+4, on non-overlapping weekly returns; partial IC controlling for the past-4-week return; tercile double sorts. (c) Weekly top/bottom-N baskets and their next-week return. (d) Earnings event windows: 10-day pre-print signals vs reaction, 15-day total and drift.
- **Caveats.** Forward windows longer than a week overlap, so their t-stats overstate independence; the lead/lag section uses non-overlapping weekly returns for that reason. Lexicon tone is not LLM-scored. The own corpus and sell-side legs cover only 2026. Non-USD names carry FX noise in raw returns.

3. Weekly panel — forward IC by horizon

Bloomberg track, 2024-01-05 → 2026-08-28 (139 weekly cross-sections). Positive IC = high score preceded outperformance. Q5-Q1 = top minus bottom quintile mean forward relative return.

Signal	IC 1w	IC 15d	IC 4w	t (4w)	hit	Q5-Q1 4w	IC 4w by year
BBG Twitter publication count vs trailing 4w	-0.004	-0.009	-0.006	-0.6	43%	+0.2%	24:-0.008 / 25:-0.023 / 26:+0.025
BBG news publication count vs trailing 4w	-0.005	-0.001	+0.011	1.1	52%	+0.5%	24:+0.006 / 25:+0.004 / 26:+0.031
BBG Twitter sentiment (daily avg, count-weighted)	+0.008	+0.016	+0.026	2.5	59%	+1.4%	24:+0.035 / 25:+0.022 / 26:+0.016
BBG Twitter sentiment change vs trailing 4w	-0.013	-0.010	-0.014	-1.7	41%	+0.5%	24:-0.032 / 25:-0.007 / 26:+0.004
BBG news sentiment (daily avg, count-weighted)	+0.006	+0.013	+0.036	3.4	59%	+1.8%	24:+0.041 / 25:+0.023 / 26:+0.048
EPS revision, BEST_EPS 1BF, 4w %	+0.039	+0.046	+0.048	3.2	68%	+3.3%	24:+0.075 / 25:+0.046 / 26:+0.004
Analyst rating drift, 4w	+0.010	+0.010	+0.019	2.0	55%	+0.7%	24:+0.044 / 25:+0.012 / 26:-0.012

Short interest % float (level)	+0.010	+0.024	+0.033	1.9	60%	+4.6%	24:-0.030 / 25:+0.070 / 26:+0.077
HOT composite	-0.004	-0.005	+0.003	0.2	43%	+0.3%	24:-0.000 / 25:-0.009 / 26:+0.027
BULL composite	+0.010	+0.022	+0.040	3.6	67%	+2.2%	24:+0.061 / 25:+0.029 / 26:+0.023

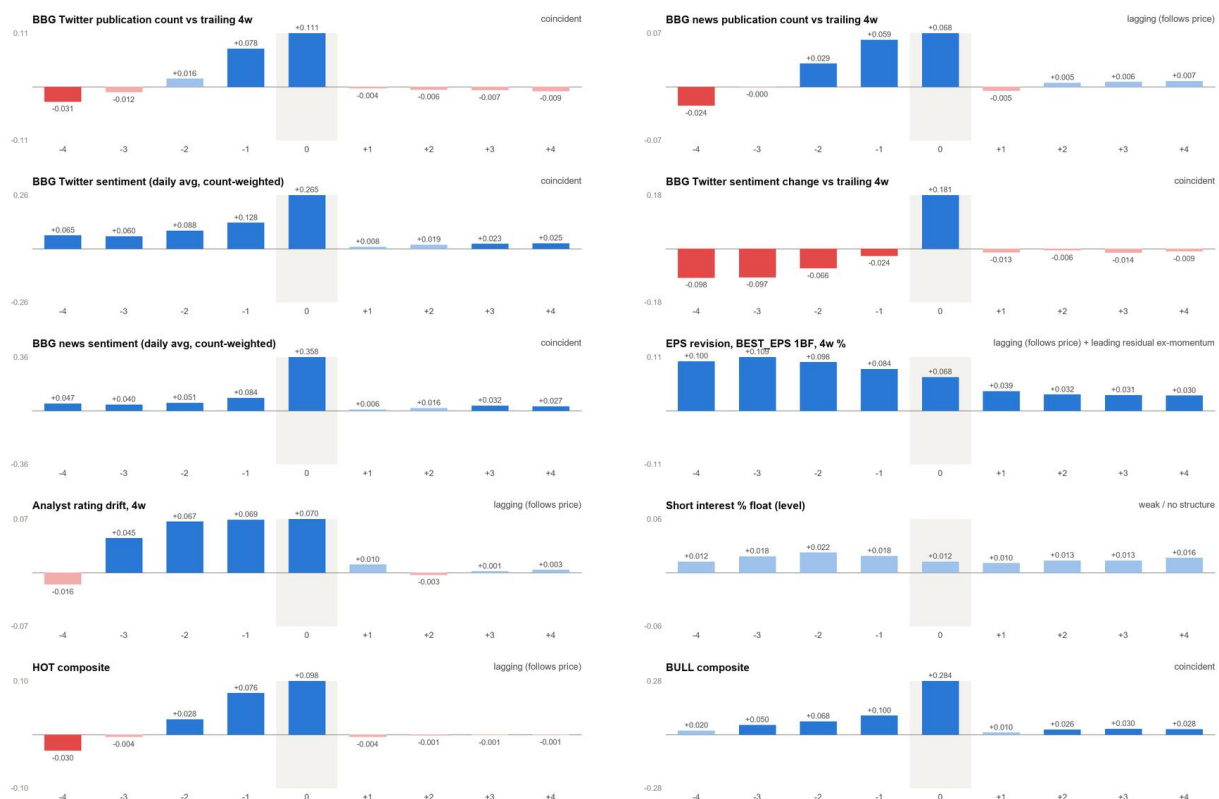
Own corpus + sell-side, 2026-05-01 → 2026-08-28 (18 weeks — indicative only).

Signal	IC 1w	IC 15d	t (15d)	hit	Q5–Q1 15d	IC 4w	weeks
Attention (curated mentions vs trailing 4w)	+0.025	+0.040	1.9	64%	+1.2%	+0.061	14
News-wire volume vs trailing 4w	-0.023	-0.027	-1.0	29%	-1.5%	-0.004	14
Tweet tone (lexicon, engagement-weighted)	-0.028	-0.041	-1.7	27%	-1.6%	-0.033	15
Tweet tone change vs trailing 4w	+0.001	-0.012	-0.5	64%	-1.5%	-0.028	14
EPS revision, BEST_EPS 1BF, 4w %	-0.049	-0.092	-1.9	12%	-2.0%	-0.111	16
Analyst rating drift, 4w	-0.045	-0.042	-1.2	31%	-1.6%	-0.019	16
Short interest % float (level)	-0.074	-0.076	-1.1	38%	-0.8%	-0.066	16
Sell-side note flow vs trailing 4w (Outlook)	-0.042	-0.068	-2.4	31%	-1.5%	-0.029	16
Sell-side subject tone (lexicon)	+0.018	-0.004	-0.1	62%	-1.0%	+0.044	16
Sell-side net rating changes, week	+0.035	+0.006	0.3	62%	+0.3%	+0.048	16
Sell-side avg price-target change %, week	+0.011	-0.002	-0.1	44%	+0.6%	+0.017	16
Sell-side net estimate revisions, week	-0.008	-0.006	-0.3	44%	+0.2%	+0.002	16
HOT composite	+0.002	+0.009	0.5	50%	-0.3%	+0.036	14
BULL composite	-0.026	-0.068	-2.0	38%	-2.7%	-0.059	16

4. Lead or lag? Cross-correlation with weekly returns

The forward-IC tables cannot separate a leading signal from sentiment that follows price. The cross-correlation function does: for a signal measured at the end of week t , the bars show its rank correlation with the relative return of week $t+k$. Bars left of the shaded column are returns that happened before the signal (sentiment reacting to price); bars to the right are returns that came after (the signal leading). Weekly returns do not overlap, so each bar's t-stat is honest.

Cross-correlation with weekly universe-relative returns, Jan-2024 → Aug-2026 (139 weeks). Bars left of the shaded k=0 column: sentiment reacting to price. Right: sentiment leading price. Faded: $|t| < 2$.



Signal	k=-2	k=-1	k=0	k=+1	k=+2	partial +1 (ex-momentum)	t	Verdict
BBG Twitter publication count vs trailing 4w	+0.016	+0.078	+0.111	-0.004	-0.006	-0.006	-0.6	coincident
BBG news publication count vs trailing 4w	+0.029	+0.059	+0.068	-0.005	+0.005	-0.006	-0.7	lagging (follows price)
BBG Twitter sentiment (daily avg, count-weighted)	+0.088	+0.128	+0.265	+0.008	+0.019	-0.002	-0.2	coincident
BBG Twitter sentiment change vs trailing 4w	-0.066	-0.024	+0.181	-0.013	-0.006	-0.012	-1.2	coincident
BBG news sentiment (daily avg, count-weighted)	+0.051	+0.084	+0.358	+0.006	+0.016	-0.002	-0.2	coincident
EPS revision, BEST_EPS 1BF, 4w %	+0.098	+0.084	+0.068	+0.039	+0.032	+0.034	2.4	lagging (follows price) + leading residual ex-momentum
Analyst rating drift, 4w	+0.067	+0.069	+0.070	+0.010	-0.003	+0.008	0.8	lagging (follows price)
Short interest % float (level)	+0.022	+0.018	+0.012	+0.010	+0.013	+0.005	0.3	weak / no structure
HOT composite	+0.028	+0.076	+0.098	-0.004	-0.001	-0.006	-0.6	lagging (follows price)
BULL composite	+0.068	+0.100	+0.284	+0.010	+0.026	+0.002	0.2	coincident

Reverse direction — this week's return → next week's signal. A large positive number is the signature of a lagging signal.

Signal	IC	t	weeks
BBG Twitter sentiment (daily avg, count-weighted)	+0.128	10.7	137
BBG Twitter publication count vs trailing 4w	+0.078	7.9	137
BBG news publication count vs trailing 4w	+0.059	5.5	137
EPS revision, BEST_EPS 1BF, 4w %	+0.084	6.0	137
Analyst rating drift, 4w	+0.069	7.2	137

Double sort — tercile of past-4-week return (rows) × tercile of the signal (columns) → mean forward 15-day relative return. If a signal only proxies momentum the columns look alike within each row; if it adds information the right column beats the left inside every row.

EPS revision, BEST_EPS 1BF, 4w %

	signal low	signal mid	signal high	high – low
past 4w lo	-0.8%	-0.8%	+0.2%	+1.0%
past 4w mid	-1.3%	-0.5%	+0.1%	+1.3%
past 4w hi	+0.4%	-0.4%	+1.1%	+0.7%

Short interest % float (level)

	signal low	signal mid	signal high	high – low
past 4w lo	-0.6%	-0.6%	+0.7%	+1.3%
past 4w mid	-1.1%	-0.5%	+0.2%	+1.3%
past 4w hi	-0.2%	+0.3%	+1.6%	+1.8%

BBG Twitter sentiment (daily avg, count-weighted)

	signal low	signal mid	signal high	high – low
past 4w lo	-0.3%	-0.5%	-0.4%	-0.1%
past 4w mid	-0.7%	-0.4%	-0.6%	+0.2%
past 4w hi	+1.1%	+0.4%	+1.0%	-0.1%

BBG Twitter publication count vs trailing 4w

	signal low	signal mid	signal high	high – low
past 4w lo	-0.3%	-0.6%	-0.3%	-0.0%
past 4w mid	-0.4%	-0.6%	-0.5%	-0.1%
past 4w hi	+0.9%	+0.9%	+0.7%	-0.3%

5. What we would have seen — weekly top/bottom-5 baskets

Every Friday, rank the universe on one signal, take the five highest and five lowest names, and record their relative return over the following week. The chart compounds the long-top / short-bottom spread; the table shows the average pick outcome per signal.

Weekly picks — cumulative spread of long top-5 / short bottom-5, equal weight, rebalanced every Friday, universe-relative, no costs



Signal (Bloomberg track)	top-5 next wk	bottom-5 next wk	spread	spread > 0	cumulative	weeks
Short interest % float (level)	+0.71%	-0.17%	+0.88%	54%	+103%	138
EPS revision, BEST_EPS 1BF, 4w %	+0.88%	+0.10%	+0.78%	55%	+124%	138
BBG Twitter sentiment (daily avg, count-weighted)	-0.05%	-0.60%	+0.54%	57%	+85%	138
BULL composite	+0.29%	-0.01%	+0.30%	55%	+29%	138
BBG Twitter publication count vs trailing 4w	-0.07%	-0.14%	+0.07%	53%	-1%	136
Analyst rating drift, 4w	+0.04%	+0.27%	-0.23%	46%	-39%	138
HOT composite	-0.15%	+0.20%	-0.35%	48%	-47%	136
BBG news publication count vs trailing 4w	+0.06%	+0.43%	-0.37%	46%	-49%	136
BBG news sentiment (daily avg, count-weighted)	-0.48%	-0.05%	-0.43%	50%	-53%	138
BBG Twitter sentiment change vs trailing 4w	-0.38%	+0.10%	-0.48%	47%	-53%	136

Own corpus + sell-side track, 2026-05-01 → 2026-08-28 — indicative only.

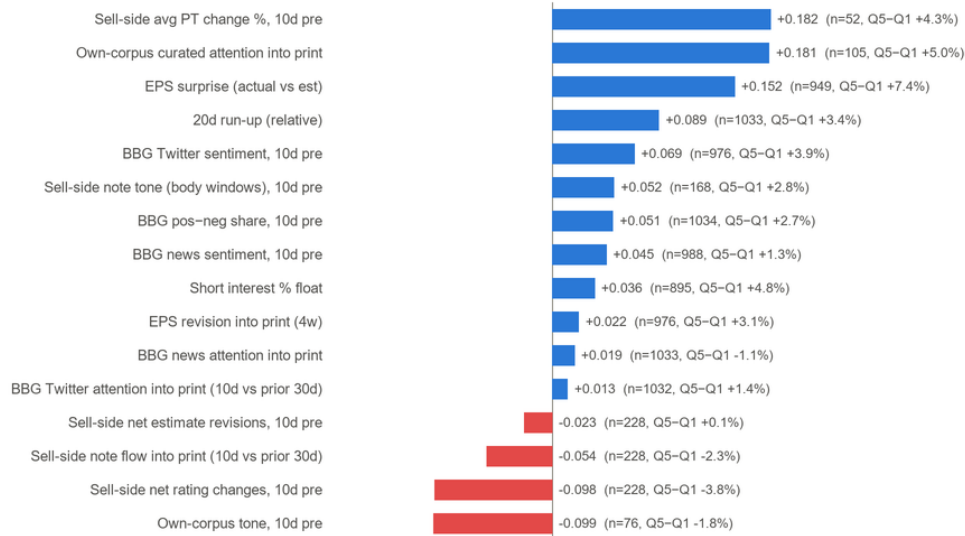
Signal (own track)	top-5 next wk	bottom-5 next wk	spread	spread > 0	cumulative	weeks
Sell-side avg price-target change %, week	+1.25%	-0.43%	+1.68%	65%	+29%	17
Sell-side note flow vs trailing 4w (Outlook)	+0.54%	-0.41%	+0.95%	65%	+15%	17
Tweet tone change vs trailing 4w	+1.22%	+0.67%	+0.55%	53%	+5%	15
Analyst rating drift, 4w	+0.04%	+0.32%	-0.28%	41%	-8%	17
Sell-side net rating changes, week	+0.28%	+0.68%	-0.40%	53%	-8%	17
HOT composite	+0.55%	+1.05%	-0.50%	47%	-9%	15
EPS revision, BEST_EPS 1BF, 4w %	-0.35%	+0.46%	-0.81%	35%	-16%	17
Tweet tone (lexicon, engagement-weighted)	-0.22%	+0.68%	-0.90%	44%	-15%	16
News-wire volume vs trailing 4w	-0.94%	+0.17%	-1.11%	47%	-18%	15
BULL composite	+0.98%	+2.36%	-1.38%	35%	-24%	17
Short interest % float (level)	-1.26%	+0.27%	-1.53%	41%	-31%	17
Attention (curated mentions vs trailing 4w)	-0.27%	+1.46%	-1.73%	33%	-25%	15
Sell-side subject tone (lexicon)	+0.07%	+2.09%	-2.02%	29%	-31%	17

Reading the tone row: the spread is positive because the most-criticised names kept falling, not because the most-praised names rose. Tone is a warning on the losers, not a buy list. The attention row is the mirror image of a lead: the hottest names underperformed the following week.

6. Earnings event windows

For every print of the 99 names (1,034 events with a full 15-day window): signals measured over the 10 calendar days before the print (baseline = the 30 days before that); outcomes from the last close before the print — reaction (to the next close), 15 days total, drift (post-print close to day +15). All relative to the equal-weight universe. Own-corpus signals exist from 2026-05-01, sell-side from 2026-01-30.

Earnings windows — pre-print signal (10 days before) → 15-day relative return after the print, rank IC across 1,034 prints (Jan-2024 → Aug-2026)



Signal	IC reaction	Q5-Q1	IC 15d	Q5-Q1	IC drift	n
BBG Twitter attention into print (10d vs prior 30d)	+0.009	-0.6%	+0.013	+1.4%	+0.014	1032
BBG Twitter sentiment, 10d pre	+0.070	+3.2%	+0.069	+3.9%	+0.028	976
BBG pos-neg share, 10d pre	+0.060	+2.5%	+0.051	+2.7%	+0.011	1034
BBG news attention into print	+0.021	-0.3%	+0.019	-1.1%	+0.002	1033
BBG news sentiment, 10d pre	+0.093	+2.6%	+0.045	+1.3%	-0.028	988
Own-corpus curated attention into print	+0.163	+5.1%	+0.181	+5.0%	+0.017	105
Own-corpus tone, 10d pre	+0.081	+3.5%	-0.099	-1.8%	-0.197	76
Sell-side note flow into print (10d vs prior 30d)	-0.030	-2.3%	-0.054	-2.3%	-0.020	228
Sell-side note tone (body windows), 10d pre	-0.022	-0.6%	+0.052	+2.8%	+0.078	168
Sell-side net rating changes, 10d pre	-0.122	-5.5%	-0.098	-3.8%	-0.009	228
Sell-side avg PT change %, 10d pre	-0.038	-0.8%	+0.182	+4.3%	+0.085	52
Sell-side net estimate revisions, 10d pre	-0.005	-1.3%	-0.023	+0.1%	+0.018	228
EPS revision into print (4w)	-0.025	+0.8%	+0.022	+3.1%	+0.065	976
20d run-up (relative)	+0.075	+1.4%	+0.089	+3.4%	+0.038	1033
Short interest % float	-0.023	+0.9%	+0.036	+4.8%	+0.055	895
EPS surprise (actual vs est)	+0.217	+7.9%	+0.152	+7.4%	-0.014	949
BBG sentiment shift, days 0..+3 minus pre	+0.424	+12.7%	+0.307	+12.2%	-0.009	967
Own-corpus tone shift, days 0..+3 minus pre	-0.032	-0.6%	+0.122	-0.5%	+0.170	68
Sell-side net rating changes, days 0..+3	+0.005	+0.1%	+0.036	+4.2%	+0.108	228
Sell-side avg PT change %, days 0..+3	+0.334	+9.8%	+0.177	+7.7%	-0.134	124

Attention into the print × result (Bloomberg Twitter attention terciles; beat/miss = EPS surprise beyond ±2%). Read across a row to see whether being hot into the print changes the payoff for the same result.

attention	beat: react	15d	n	inline: react	15d	n	miss: react	15d	n
hot	+1.4%	+0.4%	233	-3.0%	-5.0%	52	-2.5%	-1.3%	31
mid	+1.8%	+0.3%	241	-0.9%	-3.4%	37	-1.5%	+0.6%	38
cold	+2.2%	+1.7%	231	-3.5%	-4.3%	43	-3.2%	-3.3%	42

7. Implications and next steps

- **Reframe the indicator.** Present hot/cold as a crowding and positioning read (who is being talked about, who is being upgraded into a print), and bull/bear as context. Neither is a forecast of next week's tape.
- **Put the forecasting weight where the evidence is.** EPS revision momentum (small, persistent, fading in 2026), short interest as a level (squeeze regime), and the pre-print setup: sentiment and curated attention in the 10 days before a print, crossed with the result.
- **Watch the hot-into-print penalty.** Hot names that beat earned less than a quarter of what cold names that beat earned over 15 days; hot names that printed in line lost 5%. That is an actionable positioning rule ahead of the season.
- **Let the own-corpus and sell-side tracks accumulate.** Eighteen weeks is not a sample. The nightly sweep now runs (17:40 weekdays) and both tracks rebuild in the dashboard; revisit after a full earnings season.
- **Upgrades worth testing.** LLM-scored tone on the curated tweets and the sell-side bodies (the lexicon is the weakest link); an event-window version of the sell-side leg with bodies; options skew as a second crowding leg.

Files: [_wiki/_dashboards/sentiment.html](#) (interactive, rebuilt nightly) · [_wiki/_tools/build_sentiment.py](#) (steps tweets, bbg, bbglong, mail_merge, sellside, panel, long, leadlag, picks, events_pull, events, dash) · this report: [_wiki/_tools/build_sentiment_report.py](#).